

Introduction

Complex sensors deliver dozens to hundreds of channels that overlap heavily and arrive in natural groups: inertial units, excitation wavelengths, scalp regions, process subsystems.

For cost, power, bandwidth and interpretability we often want to keep a handful of the actual channels rather than a learned mixture of them.

The hard part is deciding **which** channels to keep when no class labels are available.

Problem Statement

Unsupervised filters such as variance ranking or PCA loadings score each channel by its own **marginal** statistics: fast, but blind to how channels depend on one another, and biased toward the loudest signal, which is not always the informative one.

Supervised dependency-aware selectors (mRMR, JMI) model conditional relevance but need labels.

Few methods are at once **label-free**, **dependency-aware** and **discrete**, returning real channels rather than a projection.

Novelty

A **label-free, dependency-aware selector that returns real channels**. A group-structured autoencoder with a **compressive bottleneck** learns the joint structure of all channels; **perturbing each latent factor** and measuring the per-channel reconstruction change gives a label-free relevance; a **relevance-redundancy rule** turns it into an ordered shortlist. Only the encoder changes between modalities. Its channels match a supervised selector and beat PCA and variance where the discriminative signal is not the loudest.

Method: compressive group autoencoder → perturbation attribution → relevance-redundancy selection

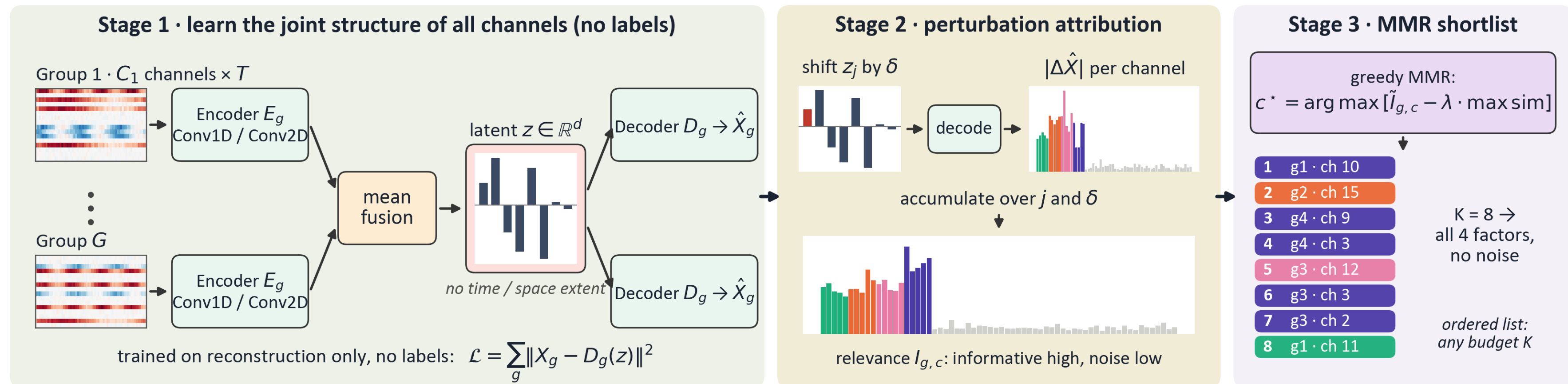


Figure 1: The three-stage pipeline with real intermediate outputs from the controlled benchmark: per-group encoders and mean fusion give a compressive code z ; shifting each latent coordinate and decoding attributes a reconstruction change to every channel; MMR orders the channels.

1 · Compressive group autoencoder

Each group is encoded independently (Conv1D for signals, Conv2D for images); the features are averaged into a latent code z with no time or space extent and decoded per group. Training uses reconstruction error only.

$$z := \frac{1}{G} \sum_{g=1}^G E_g(X_g), \quad \hat{X}_g := D_g(z), \quad \mathcal{L} := \sum_{g=1}^G \|X_g - D_g(z)\|_2^2$$

Protocol: selection is always unsupervised. Labels are used only by a downstream classifier that scores the selected K channels (accuracy or macro-F1) against variance ranking, PCA loadings, a supervised mutual-information (MI) selector and random-K.

2 · Perturbation attribution

Latent coordinates are ranked by variance and the top m kept; each is shifted by magnitudes δ , and the accumulated per-channel change of the decoded output, weighted by its variance rank w_j , is the label-free relevance.

$$I_{g,c} := \sum_{j=1}^m \sum_{\delta \in \mathcal{D}} w_j \frac{1}{NT} \sum_{n=1}^N \sum_{t=1}^T [D_g(z + \delta e_j) - D_g(z)]_{n,t,c}$$

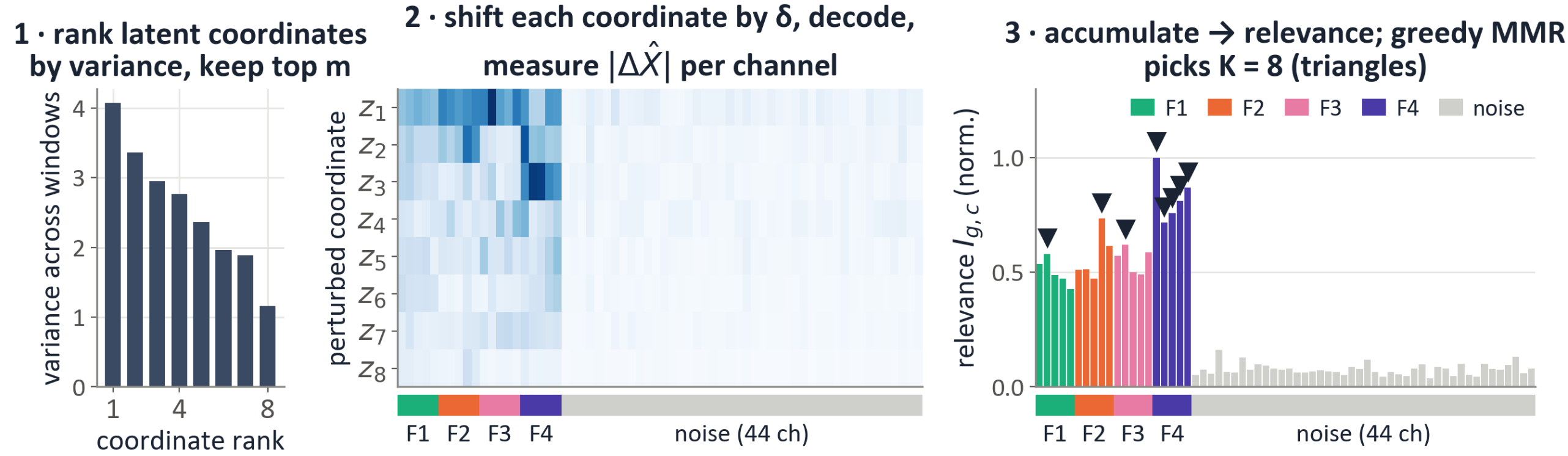


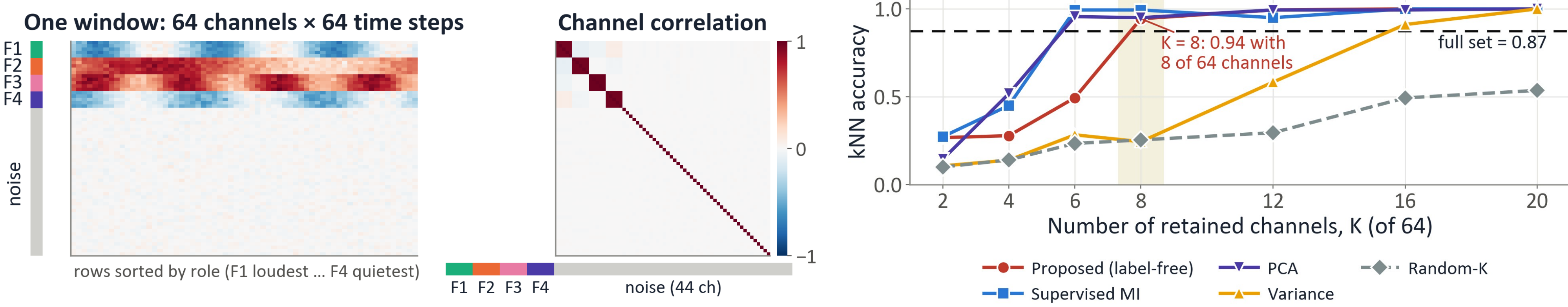
Figure 2: Attribution on the benchmark: latent coordinates ranked by variance (1), per-channel reconstruction change of each coordinate (2), accumulated relevance with the eight MMR picks (3).

3 · Relevance-redundancy selection

Relevances are normalised within each group; channels are picked greedily by maximal marginal relevance (relevance minus $\lambda \times$ cosine similarity to channels already picked). The ordered output truncates to any budget K .

$$c^* = \arg \max_{(g,c) \notin S} [\tilde{I}_{g,c} - \lambda \max_{s \in S} \text{sim}((g, c), s)], \quad \tilde{I}_{g,c} := \frac{I_{g,c}}{\max_{c'} I_{g,c'}}$$

Controlled benchmark: four latent factors hidden among 64 channels

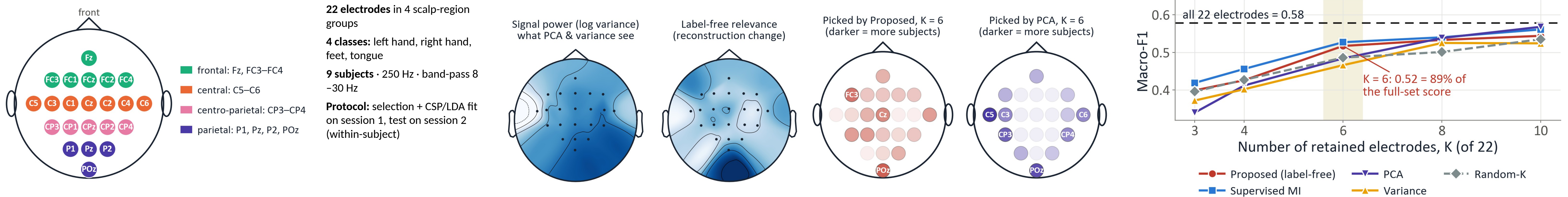


(a) 64 grouped time-series channels: four independent latent factors, each in a cluster of five correlated channels of decreasing amplitude, plus 44 pure-noise channels. The class (16 = 2^4 factor states) is decodable only by covering all four factors.

(b) kNN accuracy vs K . Eight label-free channels beat the full noisy set (0.94 vs 0.87) and match the supervised selector; PCA also recovers the factors, which here are the loudest channels; variance piles into one cluster (0.24).

(c) The channels each selector keeps at $K = 8$: the label-free method covers all four factors with no noise channel. (The bottleneck occasionally under-encodes a factor: 0.70 ± 0.29 mean accuracy over random draws of the benchmark.)

Motor-imagery EEG (BCI Competition IV-2a): the informative rhythm is not the loudest signal

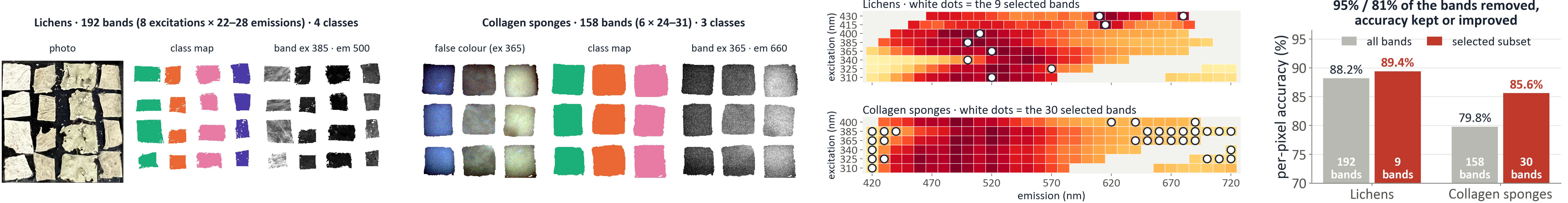


(a) Montage and groups: the grouping is acquisition metadata, not labels.

(b) Why marginal methods fail here: signal power peaks over lateral and occipital electrodes (muscle, alpha, ocular activity), so variance and PCA pick loud but uninformative electrodes (C5, C6, PO2); the reconstruction-based relevance concentrates on the sensorimotor strip (FC3, Cz, C1, CP1, CP3).

(c) Macro-F1 vs K , mean over nine subjects. At six of 22 electrodes the label-free method reaches 0.517 (89% of the full 0.578), above PCA, variance and random and close to supervised MI.

Biomedical multi-excitation fluorescence imaging: up to 95% of the bands removed without loss

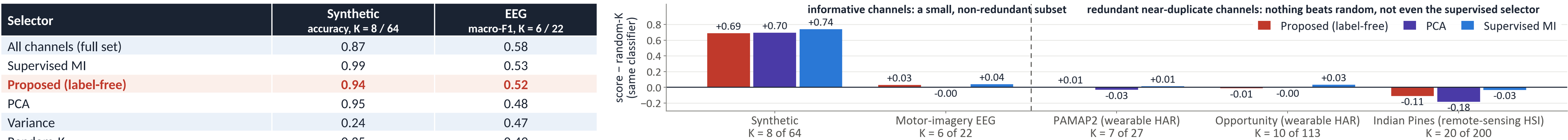


(a) Multi-excitation fluorescence cubes: for each excitation wavelength a full emission spectrum is recorded per pixel, so every excitation-emission pair is one image and one channel: 192 bands for lichens (4 classes), 158 for collagen sponges (3 classes). The band images shown are among the selected ones.

(b) Where the selected bands sit on the excitation $\times$ emission grid (background: mean emission per excitation, row-normalised). The picks spread across excitations instead of clustering on the brightest bands.

(c) 9 of 192 lichen bands: 89.4% vs 88.2% with all bands (a 95% reduction); 30 of 158 collagen bands: 85.6% vs 79.8%. Pruning also denoises.

Results across regimes: where label-free selection pays, and where nothing beats a random subset



Conclusion and Next Steps

- A compressive group autoencoder, perturbation attribution and a relevance-redundancy rule select real channels without labels; only the encoder changes between modalities.
 - The selected channels match a supervised selector and beat PCA and variance where the discriminative signal is not the loudest; up to 95% of imaging bands removed without loss.
 - A simple label-using check tells in advance when any selection is pointless.
- Next:** tighten the bottleneck, make the attribution less sensitive to channel variance, scale to larger channel arrays.

References

- Balin, Abid, Zou. Concrete autoencoders: differentiable feature selection and reconstruction. ICML 2019.
- Peng, Long, Ding. Feature selection based on mutual information (mRMR). IEEE TPAMI 27(8), 2005.
- Brown, Pocock, Zhao, Luján. Conditional likelihood maximisation (JMI). JMLR 13, 2012.
- He, Cai, Niyogi. Laplacian score for feature selection. NeurIPS 2005.
- Cai, Liu, Cai. BS-Nets: end-to-end band selection of hyperspectral images. IEEE TGRS 58(3), 2020.
- Rajabinasab, Houle, Chelly, Zimek. Worse than random? A baseline for unsupervised feature selection. arXiv:2605.22973, 2026.
- Tangermann et al. Review of the BCI Competition IV. Front. Neurosci. 6, 2012.

Acknowledgments

Financial support from the European Union NAS SAR-101087403 and the United States R44 HL120511 awards is gratefully acknowledged. Narek Chilingaryan and Kristina Ghahramanyan are thanked for providing the labelled lichen and collagen-sponge datasets.

Paper: DOI 10.1109/CODASSA69992.2026.00059 **Code:** github.com/Nard248/spectral-select

